POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5051 | Semester V | Theory | 4 Credits

# Automobile Chassis and Climate Control

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

| | |
|---|---|
| Programmes | **Automobile Engineering** |
| Contact hours | **60** |
| Teaching scheme | **3:1:0:0** |
| Credits | **4** |
| CIE / ESE | **Theory CIE 40 / Theory ESE 60** |
| Self-learning | **5.5 h/week** |

CURRICULUM INTEGRITY

## Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5051 as Automobile Chassis and Climate Control in Semester V for Automobile Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on typical polytechnic automobile engineering curricula and automotive HVAC standards. It does not claim to reproduce official REV2026 detailed-syllabus wording.

**Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Chassis repairs, brake servicing and refrigerant handling must be performed by certified technicians using approved equipment, manufacturer data and current environmental/safety regulations.

**Verified source note:** Verified sources support chassis components (frames, axles, steering, brakes, suspension) and automotive climate control (refrigeration cycle, HVAC components). The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

## What You Are Expected to Learn

**60**
Contact hours

**4**
Credits

**Theory CIE 40**
CIE

**Theory ESE 60**
ESE

### Course introduction

Automobile Chassis and Climate Control develops the principles and maintenance practices for vehicle structural frameworks, suspension, steering, braking systems and passenger-comfort systems. It relates chassis geometry and dynamics to safety, stability and comfort, while covering the thermodynamics and control of automotive HVAC systems.

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### Prerequisite foundation

Automobile engineering fundamentals, basic thermodynamics, fluid mechanics and engineering mechanics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

**Teaching and assessment scheme**

| L | T | P | S | Self-learning/week | Contact hours | Credits | CIE | ESE |
|---|---|---|---|---|---|---|---|---|
| 3 | 1 | 0 | 0 | 5.5 h/week | 60 | 4 | Theory CIE 40 | Theory ESE 60 |

STUDY METHOD

## How to Use This Handbook

### 1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

### 2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

### 3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

### 4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

## Objectives and Course Outcomes

### Official course objectives

1. Explain the construction and types of automobile frames, front axles and steering systems.
2. Describe the principles and components of automotive suspension and braking systems.
3. Explain the working principles of wheels, tyres and automotive climate control (HVAC) systems.
4. Describe the environmental and safety regulations related to chassis and climate control systems.

### Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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| Official Course Outcomes | | |
|---|---|---|
| **CO** | **Official outcome** | **Bloom level** |
| CO1 | Explain the construction of vehicle frames, front axles and steering geometry. | Understand |
| CO2 | Describe the working of various suspension and braking systems for safety and comfort. | Understand |
| CO3 | Explain the principles of automotive air conditioning and heating systems. | Understand |
| CO4 | Apply maintenance and troubleshooting procedures for chassis and climate control components. | Apply |

| Official CO-PO articulation matrix | | | | | | | |
|---|---|---|---|---|---|---|---|
| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7-PO11 |
| CO1 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO2 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO3 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO4 | 3 | 2 | 2 | 2 | 2 | – | – |

SYLLABUS COVERAGE

# Curriculum Coverage Map

**All official major topics mapped to handbook chapters**

| Module | Title | Official major topics | Hours | CO | Handbook section |
|---|---|---|---|---|---|
| 1 | Frames, Front Axles and Steering | Types of chassis and frames; front axle construction; steering geometry (caster, camber, toe); steering systems and linkages; power steering basics. | Approx. 14 hours | CO1 | Module 1 |
| 2 | Suspension and Braking Systems | Purpose of suspension; types (Rigid, Independent); springs and shock absorbers; braking principles; Hydraulic, Air and ABS systems; brake components (Disc, Drum). | Approx. 16 hours | CO2 | Module 2 |
| 3 | Wheels, Tyres and Climate Control | Wheel types and construction; Tyre construction (Radial, Bias); tyre ratings; fundamentals of automotive HVAC; Refrigeration cycle; AC components. | Approx. 16 hours | CO3 | Module 3 |
| 4 | HVAC Control and Maintenance | Heating and ventilation systems; AC control circuits; sensors and actuators; troubleshooting chassis and AC systems; environmental regulations (refrigerant handling). | Approx. 14 hours | CO4 | Module 4 |

LEARNING ROADMAP

# How the Modules Connect

### Frames, Front Axles and Steering

Types of chassis and frames; front axle construction; steering geometry (caster, camber, toe); steering systems and linkages; power steering basics.

CO1 · Approx. 14 hours

### Suspension and Braking Systems

Purpose of suspension; types (Rigid, Independent); springs and shock absorbers; braking principles; Hydraulic, Air and ABS systems; brake components (Disc, Drum).

CO2 · Approx. 16 hours

### Wheels, Tyres and Climate Control

Wheel types and construction; Tyre construction (Radial, Bias); tyre ratings; fundamentals of automotive HVAC; Refrigeration cycle; AC components.

CO3 · Approx. 16 hours

### HVAC Control and Maintenance

Heating and ventilation systems; AC control circuits; sensors and actuators; troubleshooting chassis and AC systems; environmental regulations (refrigerant handling).

CO4 · Approx. 14 hours

**MODULE 1**

## Frames, Front Axles and Steering

Types of chassis and frames; front axle construction; steering geometry (caster, camber, toe); steering systems and linkages; power steering basics.

**Approx. 14 hours** **CO1** **Understand · Apply**

### Learning goals

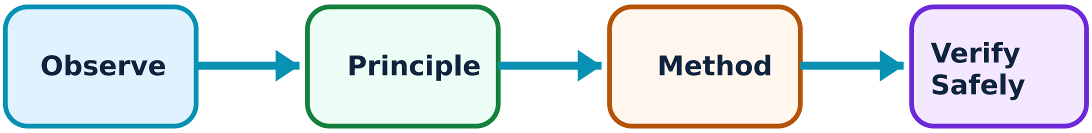
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Frames, Front Axles and Steering: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Vehicle frames and front axles−

The chassis frame supports the vehicle's weight and components. Frames can be conventional, semi-integral or unit (monocoque) construction. Front axles support the weight and provide steering pivots, classified into 'Live' and 'Dead' axles.

### Study and application method

Compare different types of vehicle frames in a conceptual table; describe the construction of an 'Elliot' type front axle.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Compare different types of vehicle frames in a conceptual table; describe the construction of an 'Elliot' type front axle.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Frame damage affects vehicle alignment and safety. Do not attempt frame repairs without authorised alignment jigs and structural integrity verification.

## 2. Steering geometry and systems−

Steering geometry ensures stability and reduces tyre wear. Key parameters include Caster, Camber, Kingpin Inclination and Toe-in/Toe-out. Steering systems (e.g., Rack and Pinion, Recirculating Ball) convert steering wheel rotation into wheel movement.

### Study and application method

Define 'Caster' and 'Camber' and explain their effects on steering; sketch a typical steering linkage for a front-engine vehicle.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Define 'Caster' and 'Camber' and explain their effects on steering; sketch a typical steering linkage for a front-engine vehicle.

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**Common mistake / precaution:** Incorrect steering geometry leads to poor handling and rapid tyre wear. Use only authorised wheel-alignment equipment and factory specifications for adjustments.

**Module 1 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

## Suspension and Braking Systems

Purpose of suspension; types (Rigid, Independent); springs and shock absorbers; braking principles; Hydraulic, Air and ABS systems; brake components (Disc, Drum).

**Approx. 16 hours** **CO2** **Understand · Apply**

### Learning goals

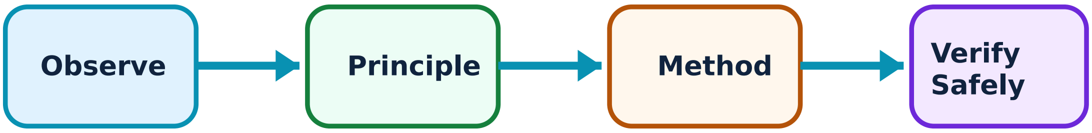
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Suspension and Braking Systems: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Suspension principles and types−

Suspension isolates the vehicle from road shocks and maintains tyre contact. Rigid axle suspension is used for heavy loads, while independent suspension (e.g., MacPherson strut) improves comfort. Springs (Leaf, Coil, Torsion) and shock absorbers manage energy.

#### Study and application method

Compare 'Rigid' and 'Independent' suspension systems conceptually; describe the working of a telescopic shock absorber.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Compare 'Rigid' and 'Independent' suspension systems conceptually; describe the working of a telescopic shock absorber.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Suspension components are under high tension. Do not dismantle struts or leaf springs without authorised spring compressors and safety precautions.

## 2. Braking systems and safety−

Brakes convert kinetic energy into heat to slow or stop the vehicle. Hydraulic brakes use Pascal's law. Air brakes are common in heavy vehicles. Anti-lock Braking Systems (ABS) prevent wheel lock-up during emergency braking to maintain steering control.

### Study and application method

Explain the working principle of a 'Hydraulic Braking' system; describe the role of the 'Wheel Speed Sensor' in an ABS system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Explain the working principle of a 'Hydraulic Braking' system; describe the role of the 'Wheel Speed Sensor' in an ABS system.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Braking is a life-critical system. Use only authorised brake fluids and genuine replacement parts. All brake work must follow authorised bleeding and testing procedures.

**Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

## Wheels, Tyres and Climate Control

Wheel types and construction; Tyre construction (Radial, Bias); tyre ratings; fundamentals of automotive HVAC; Refrigeration cycle; AC components.

**Approx. 16 hours** **CO3** **Understand · Apply**

### Learning goals

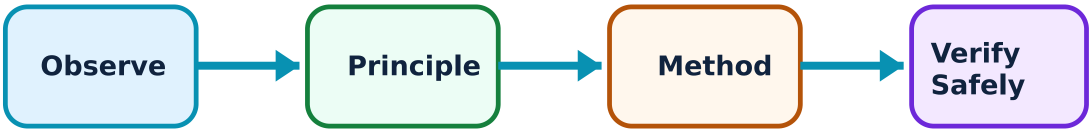
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

**Engineering habit**

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Wheels, Tyres and Climate Control: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Wheels and tyres−

Wheels support the tyres and transfer torque. Tyres provide traction and cushioning. Construction types include Bias-ply and Radial. Tyre ratings (size, load, speed) are critical for safe operation. Tubeless tyres offer better safety against sudden deflation.

### Study and application method

Describe the layers of a 'Radial' tyre conceptually; explain the meaning of tyre markings (e.g., 205/55 R16).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Describe the layers of a 'Radial' tyre conceptually; explain the meaning of tyre markings (e.g., 205/55 R16).

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Incorrect tyre pressure or worn treads increase accident risk. Follow authorised inflation charts and replace tyres when they reach the wear indicators.

## 2. Automobile air conditioning−

Automotive AC uses the vapour compression refrigeration cycle. Key components include the Compressor (driven by the engine), Condenser, Expansion Valve and Evaporator. Refrigerants (like R134a or R1234yf) transfer heat from the cabin to the outside air.

### Study and application method

Sketch the 'Vapour Compression Cycle' for an automotive AC system; list the functions of the 'Receiver-Drier'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Sketch the 'Vapour Compression Cycle' for an automotive AC system; list the functions of the 'Receiver-Drier'.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** AC refrigerants are environmental hazards and under high pressure. Do not vent refrigerants to the atmosphere; use authorised recovery and recharging equipment.

**Module 3 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

# HVAC Control and Maintenance

Heating and ventilation systems; AC control circuits; sensors and actuators; troubleshooting chassis and AC systems; environmental regulations (refrigerant handling).

**Approx. 14 hours** **CO4** **Understand · Apply**

### Learning goals

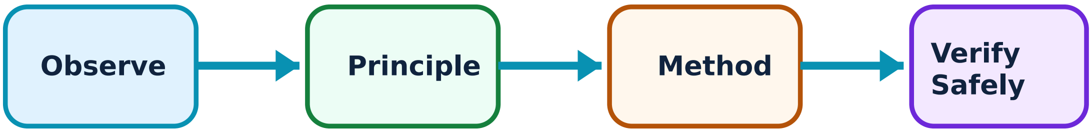
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for HVAC Control and Maintenance: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Heating and ventilation−

Heating uses engine coolant passed through a heater core. Ventilation systems use blowers and ducts to distribute air. Modern systems use Electronic Climate Control (ECC) with sensors (sun, cabin, ambient) to automatically maintain the desired temperature.

### Study and application method

Describe how heat is transferred from the engine to the cabin in a vehicle heater; explain the role of 'Blend Doors' in an HVAC system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Describe how heat is transferred from the engine to the cabin in a vehicle heater; explain the role of 'Blend Doors' in an HVAC system.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Hot engine coolant can cause severe burns. Do not open the cooling system or heater hoses when the engine is hot.

## 2. Troubleshooting and regulations−

Maintenance involves regular inspection of chassis components and AC performance. Troubleshooting identifies leaks, noises or poor cooling. Environmental regulations strictly control the handling and disposal of refrigerants to prevent ozone depletion and global warming.

### Study and application method

Develop a troubleshooting checklist for a 'Poor Cooling' complaint in an automotive AC; list the safety precautions for handling automotive refrigerants.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Develop a troubleshooting checklist for a 'Poor Cooling' complaint in an automotive AC; list the safety precautions for handling automotive refrigerants.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Handling refrigerants requires certification in many regions. All AC servicing must follow authorised environmental and safety protocols.

**Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

## Rapid Recall Bank

### Recall 1

State symbols and SI units before substitution.

### Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

### Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

### Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

### Recall 5

For a comparison: use construction, working, result, advantage and limitation.

### Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

## Diagram Library for Rapid Revision

### Module 1: Frames, Front Axles and Steering

Use the concept flow to revise observation, principle, method and verification.

### Module 2: Suspension and Braking Systems

Use the concept flow to revise observation, principle, method and verification.

### Module 3: Wheels, Tyres and Climate Control

Use the concept flow to revise observation, principle, method and verification.

### Module 4: HVAC Control and Maintenance

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

## Official Self-Learning and Student Activities

### Structured activity

Compare Leaf Spring and Coil Spring suspension in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Sketch the layout of a power-assisted steering system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Draw the block diagram of an automotive HVAC system showing the flow of refrigerant and air.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Identify the type of suspension used in a passenger car and a heavy truck in your locality.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Perform a conceptual wheel-alignment check procedure listing the steps and tools required.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

**Activity discipline:** The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

## Assessment Methodology and Examination Pattern

**Official assessment structure**

| Component | Official mark / conversion | Student evidence |
| --- | --- | --- |
| Attendance | 5 marks | End-of-semester attendance component. |
| Self-learning activities | 15 marks | Module-linked activities. |
| Series examinations | 20 + 20 converted | Two 50-mark series examinations. |
| CIE total | Theory CIE 40 | Continuous internal evaluation. |
| Written ESE | Theory ESE 60 | 2.5 hours; official Part A / Part B / Part C pattern. |

### Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

### Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

## Module-wise Questions with Answers

## Module 1 — Frames, Front Axles and Steering

**Explain Vehicle frames and front axles and state one correct method, application or precaution.−**

The chassis frame supports the vehicle's weight and components. Frames can be conventional, semi-integral or unit (monocoque) construction. Front axles support the weight and provide steering pivots, classified into 'Live' and 'Dead' axles.

**Answer framework:** Compare different types of vehicle frames in a conceptual table; describe the construction of an 'Elliot' type front axle.

**Important point:** Frame damage affects vehicle alignment and safety. Do not attempt frame repairs without authorised alignment jigs and structural integrity verification.

**Explain Steering geometry and systems and state one correct method, application or precaution.−**

Steering geometry ensures stability and reduces tyre wear. Key parameters include Caster, Camber, Kingpin Inclination and Toe-in/Toe-out. Steering systems (e.g., Rack and Pinion, Recirculating Ball) convert steering wheel rotation into wheel movement.

**Answer framework:** Define 'Caster' and 'Camber' and explain their effects on steering; sketch a typical steering linkage for a front-engine vehicle.

**Important point:** Incorrect steering geometry leads to poor handling and rapid tyre wear. Use only authorised wheel-alignment equipment and factory specifications for adjustments.

## Module 2 — Suspension and Braking Systems

**Explain Suspension principles and types and state one correct method, application or precaution.−**

Suspension isolates the vehicle from road shocks and maintains tyre contact. Rigid axle suspension is used for heavy loads, while independent suspension (e.g., MacPherson strut) improves comfort. Springs (Leaf, Coil, Torsion) and shock absorbers manage energy.

**Answer framework:** Compare 'Rigid' and 'Independent' suspension systems conceptually; describe the working of a telescopic shock absorber.

**Important point:** Suspension components are under high tension. Do not dismantle struts or leaf springs without authorised spring compressors and safety precautions.

**Explain Braking systems and safety and state one correct method, application or**

precaution.−

Brakes convert kinetic energy into heat to slow or stop the vehicle. Hydraulic brakes use Pascal's law. Air brakes are common in heavy vehicles. Anti-lock Braking Systems (ABS) prevent wheel lock-up during emergency braking to maintain steering control.

**Answer framework:** Explain the working principle of a 'Hydraulic Braking' system; describe the role of the 'Wheel Speed Sensor' in an ABS system.

**Important point:** Braking is a life-critical system. Use only authorised brake fluids and genuine replacement parts. All brake work must follow authorised bleeding and testing procedures.

## Module 3 — Wheels, Tyres and Climate Control

**Explain Wheels and tyres and state one correct method, application or precaution.−**

Wheels support the tyres and transfer torque. Tyres provide traction and cushioning. Construction types include Bias-ply and Radial. Tyre ratings (size, load, speed) are critical for safe operation. Tubeless tyres offer better safety against sudden deflation.

**Answer framework:** Describe the layers of a 'Radial' tyre conceptually; explain the meaning of tyre markings (e.g., 205/55 R16).

**Important point:** Incorrect tyre pressure or worn treads increase accident risk. Follow authorised inflation charts and replace tyres when they reach the wear indicators.

**Explain Automobile air conditioning and state one correct method, application or precaution.−**

Automotive AC uses the vapour compression refrigeration cycle. Key components include the Compressor (driven by the engine), Condenser, Expansion Valve and Evaporator. Refrigerants (like R134a or R1234yf) transfer heat from the cabin to the outside air.

**Answer framework:** Sketch the 'Vapour Compression Cycle' for an automotive AC system; list the functions of the 'Receiver-Drier'.

**Important point:** AC refrigerants are environmental hazards and under high pressure. Do not vent refrigerants to the atmosphere; use authorised recovery and recharging equipment.

## Module 4 — HVAC Control and Maintenance

**Explain Heating and ventilation and state one correct method, application or precaution.−**

Heating uses engine coolant passed through a heater core. Ventilation systems use blowers and ducts to distribute air. Modern systems use Electronic Climate Control (ECC) with sensors (sun, cabin, ambient) to automatically maintain the desired temperature.

**Answer framework:** Describe how heat is transferred from the engine to the cabin in a vehicle heater; explain the role of 'Blend Doors' in an HVAC system.

**Important point:** Hot engine coolant can cause severe burns. Do not open the cooling system or heater hoses when the engine is hot.

**Explain Troubleshooting and regulations and state one correct method, application or precaution.−**

Maintenance involves regular inspection of chassis components and AC performance. Troubleshooting identifies leaks, noises or poor cooling. Environmental regulations strictly control the handling and disposal of refrigerants to prevent ozone depletion and global warming.

**Answer framework:** Develop a troubleshooting checklist for a 'Poor Cooling' complaint in an automotive AC; list the safety precautions for handling automotive refrigerants.

**Important point:** Handling refrigerants requires certification in many regions. All AC servicing must follow authorised environmental and safety protocols.

PRACTICE ONLY

# Practice Model Paper

**Important:** This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

### Part A — short response

1. State one key definition from Module 1: Frames, Front Axles and Steering.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Suspension and Braking Systems.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Wheels, Tyres and Climate Control.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: HVAC Control and Maintenance.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

### Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

### Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Types of chassis and frames; front axle construction; steering geometry (caster, camber, toe); steering systems and linkages; power steering basics..
2. Develop a complete answer or practical plan for: Purpose of suspension; types (Rigid, Independent); springs and shock absorbers; braking principles; Hydraulic, Air and ABS systems; brake components (Disc, Drum)..
3. Develop a complete answer or practical plan for: Wheel types and construction; Tyre construction (Radial, Bias); tyre ratings; fundamentals of automotive HVAC; Refrigeration cycle; AC components..
4. Develop a complete answer or practical plan for: Heating and ventilation systems; AC control circuits; sensors and actuators; troubleshooting chassis and AC systems; environmental regulations (refrigerant handling)..

LAST-MILE REVISION

## Quick Revision

### Module 1: Frames, Front Axles and Steering

Types of chassis and frames; front axle construction; steering geometry (caster, camber, toe); steering systems and linkages; power steering basics.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 2: Suspension and Braking Systems

Purpose of suspension; types (Rigid, Independent); springs and shock absorbers; braking principles; Hydraulic, Air and ABS systems; brake components (Disc, Drum).

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 3: Wheels, Tyres and Climate Control

Wheel types and construction; Tyre construction (Radial, Bias); tyre ratings; fundamentals of automotive HVAC; Refrigeration cycle; AC components.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 4: HVAC Control and Maintenance

Heating and ventilation systems; AC control circuits; sensors and actuators; troubleshooting chassis and AC systems; environmental regulations (refrigerant handling).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

**Common mistakes:** Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

**Exam checklist:** Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

## Glossary

| Important terms from the syllabus | |
|---|---|
| **Term / topic** | **One-line meaning** |
| Vehicle frames and front axles | The chassis frame supports the vehicle's weight and components. |
| Steering geometry and systems | Steering geometry ensures stability and reduces tyre wear. |
| Suspension principles and types | Suspension isolates the vehicle from road shocks and maintains tyre contact. |
| Braking systems and safety | Brakes convert kinetic energy into heat to slow or stop the vehicle. |
| Wheels and tyres | Wheels support the tyres and transfer torque. |
| Automobile air conditioning | Automotive AC uses the vapour compression refrigeration cycle. |
| Heating and ventilation | Heating uses engine coolant passed through a heater core. |
| Troubleshooting and regulations | Maintenance involves regular inspection of chassis components and AC performance. |

LEARNING RESOURCES

# References

### Recommended automobile-chassis references


1. Kirpal Singh, Automobile Engineering (Vol. I & II).
2. K. K. Jain and R. B. Asthana, Automobile Engineering.
3. William H. Crouse and Donald L. Anglin, Automotive Mechanics.
4. Steven Daly, Automotive Air Conditioning and Climate Control Systems.


### Approved public automotive-engineering sources

- https://nptel.ac.in/courses/107106088
- https://www.sciencedirect.com/topics/engineering/automobile-chassis

These sources provide the verified study basis while official Course 5051 detail is unavailable; they do not replace official vehicle manuals, authorised service procedures or certified environmental protocols for refrigerant handling.